

Ayush Kumar

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EDUCATION

Birla Institute of Technology, Mesra

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.38)

2023 – 2027

Ranchi, Jharkhand

St. Karen's Secondary School

Class XII (Percentage: 87.2%)

2021 – 2022

Patna, Bihar

St. Karen's Secondary School

Class X (Percentage: 96.8%)

2019 – 2020

Patna, Bihar

EXPERIENCE

Research Intern — Dept. of Computer Science & Engineering, BIT Mesra

May 2026 – Jul 2026

Lunar Crater Detection from Chandrayaan-2 Imagery (YOLOv11 + ResNet101)

[\[Live Demo\]](#) | [\[GitHub\]](#)

- Trained a crater detector for OHRC and TMC-2 imagery by replacing YOLOv11's backbone with an ImageNet-pretrained ResNet101 (94M params) via a patched Ultralytics build, achieving **0.683 and 0.589 mAP50** — vs reference 0.667 / 0.587.
- Ran a backbone ablation against stock YOLOv11n: pretrained backbone roughly **doubles mAP50 and triples mAP50-95**; added a parameter-count assertion to catch a silent config-fallback failure that trained the wrong 2.6M-param model.
- Profiled CPU deployment with ONNX Runtime (20-run benchmarks): FP32 export ran **0.73–0.86×** PyTorch speed while dynamic INT8 was **15–20×** slower; root-caused the dequantize fallback — fix: static QDQ quantization.
- Deployed the model as a public Streamlit demo with weights hosted on Hugging Face Hub.

PROJECTS

CodeArena — Full-Stack Competitive Programming Judge & Contest Platform

[\[Live\]](#) | [\[GitHub\]](#)

Next.js, TypeScript, Node.js, MongoDB, Redis, BullMQ, Socket.io, Docker

Jun 2026 – Jul 2026

- Built and deployed an online judge supporting ICPC-style rated contests, practice problems, and live leaderboards on an Azure VM (Docker Compose, Caddy with automatic TLS), serving **200 users and 550+ submissions** across 3 contests.
- Engineered a secure asynchronous code-execution pipeline — submissions judged in isolated Docker sandboxes (CPU / memory / wall-time limits, no network) via BullMQ with retries and idempotent verdict writes; cut C++ compile time **~70%** by precompiling bits/stdc++.h.
- Load-tested the judge with a 15-bot harness: sustained **19.1 verdicts/min** under 45 concurrent submissions on 2 vCPUs with submit latency flat at **416ms p95**, cross-verified via Little's law.
- Stress-tested the read path with k6 to **1,000 virtual users**: 500 VUs clean (**0% errors**, p95 ≤ 610ms incl. SSR), APIs flat at **~140ms p95** through 1,000 VUs; diagnosed and fixed a reverse-proxy OOM crash-loop via kernel logs, doubling clean capacity.
- Delivered real-time verdict streaming and live leaderboards with Socket.io and Redis sorted sets, implementing ICPC penalty scoring with **O(log N)** updates and rank queries.
- Designed an AI hint system (Gemini 2.5 Flash) with submission-context injection, tiered levels to prevent solution leakage, and automatic disabling during rated contests.

Nakalchi — Source-Code Plagiarism Detection Engine & Service

[\[Live\]](#) | [\[GitHub\]](#)

TypeScript, Node.js, Express, MongoDB, Redis, BullMQ, Next.js, Docker

Jul 2026

- Implemented the MOSS winnowing algorithm (Stanford, SIGMOD '03) — hand-written C++/Python lexers, overflow-safe Rabin-Karp double-hash (**4.5×** faster than BigInt), O(n) sliding-window fingerprint selection, and property-based tests verifying the (w+k-1) detection guarantee over **10,000 trials**.
- Detected **6/6** disguised plagiarism variants (renames, rewrites, reordering, partial copies) with **0/10 false positives** via inverted-index candidate generation with shared-template elimination — **1,000 submissions analyzed in 1.42s** on a 2-vCPU VM, **~51×** fewer comparisons than naive O(n²) (9,785 vs 499,500 pairs).
- Built a crash-safe async analysis service (Express + BullMQ + MongoDB) integrated with a live judge — HMAC-signed webhooks, **exactly-once processing** proved by SIGKILLing a worker mid-analysis (identical **7,080 pairs** after restart), judge operational with the service down — and a Next.js report UI with byte-accurate side-by-side match highlighting.

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript

Frameworks & Libraries: Next.js, React, Node.js, Express, Socket.io, Tailwind CSS

Databases & Infrastructure: MongoDB, Redis, BullMQ, Docker, Caddy, Linux, Azure VM

Tools: Git, GitHub, GitHub Actions (CI), Load Testing (k6)

CODING PROFILES

- Codeforces: Pupil (1280) [\[Link\]](#)
- LeetCode: Knight (1942) [\[Link\]](#)
- CodeChef: 3-Star (1626) [\[Link\]](#)

ACHIEVEMENTS

- Solved **1300+ problems** across multiple online judges and platforms. [\[Link\]](#)
- Attained **Global Rank 307** in CodeChef Starters 246. [\[Link\]](#)
- Attained **Global Rank 673** among **37,000+** participants in LeetCode Weekly Contest 506. [\[Link\]](#)
- Accomplished **Global Rank 1696** among **34,000+** participants in Codeforces Round 1109 (Div. 3). [\[Link\]](#)
- Semi-finalist** in Flipkart GRiD 7.0 [\[Link\]](#)

POSITION OF RESPONSIBILITY

- NSS Volunteer — led an education drive for **50+ underprivileged children** spread across multiple villages.